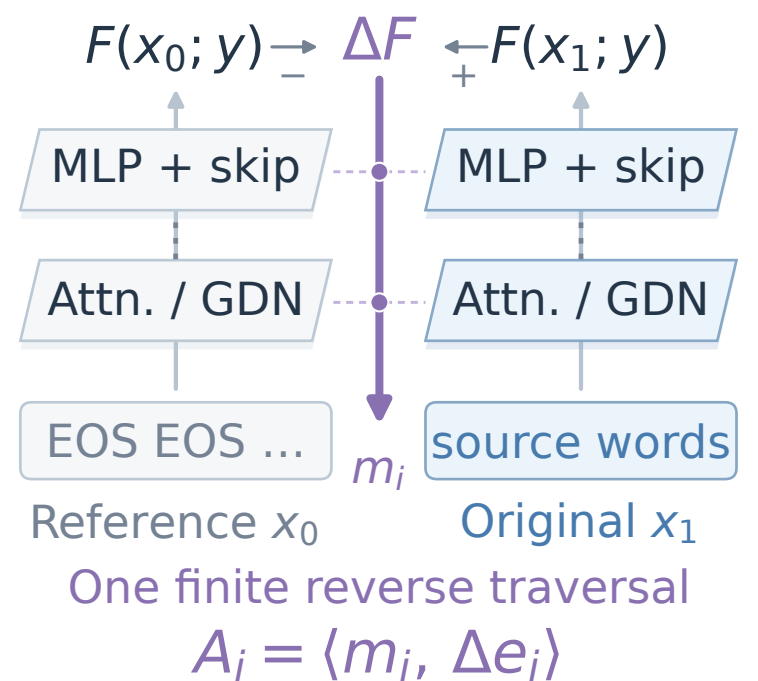


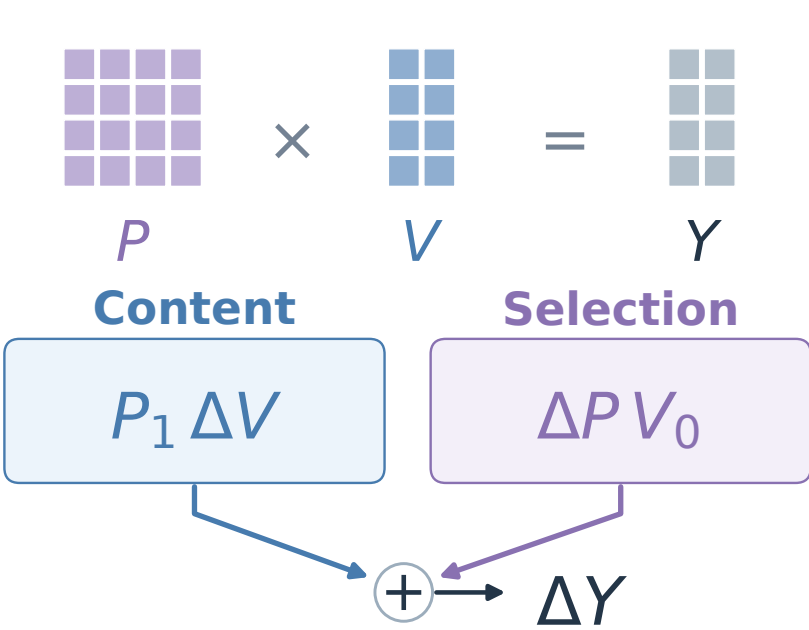
(a) Finite reverse trace

Two inputs. One fixed response y.



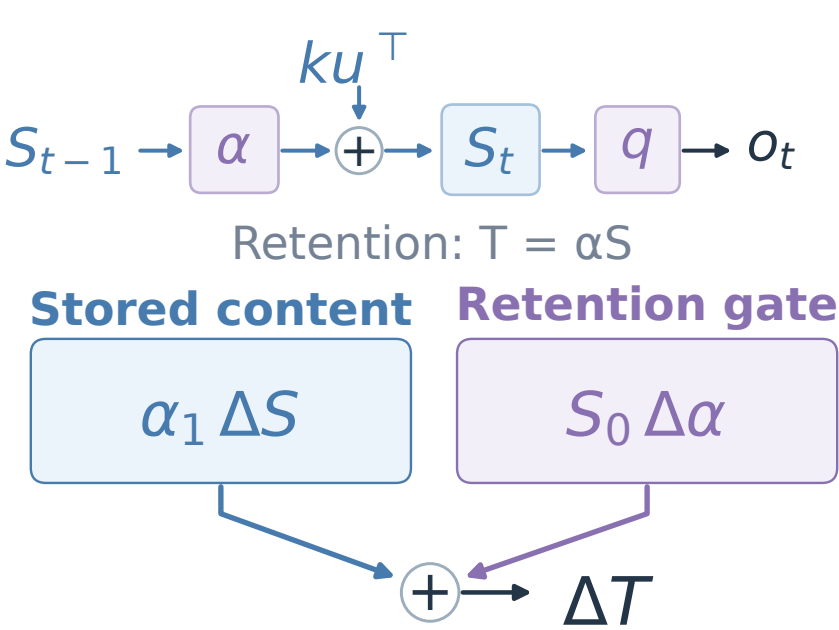
(b) Attention

Select and carry



(c) Gated DeltaNet

Retain, write, and read



(d) Choose the playwright, not the composer.

Qwen3.5 · MoreHopQA example 1

CONTEXT EXCERPTS

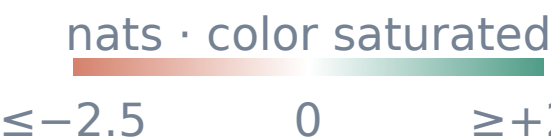
... William Shakespeare's play
... The score is by William Walton.

Name-span total (nats)

+1.56
-1.13

REQUESTED ROLE (EXCERPT)

... the author of the play ...



Full input

Shown spans underlined

Answer

William Shakespeare

$$\sum_i A_i = \Delta F$$

Across all input sources

(a) Qwen3-8B

Context

One of the special magic numbers for deafening-opium is: 5443951.

...

One of the special magic numbers for mammoth-barber is: 8698256.

Question

What are all the special magic numbers for deafening-opium, and mammoth-barber mentioned in the provided text?

Full input

**(b) Qwen3.5-9B**

Context

One of the special magic numbers for deafening-opium is: 5443951.

■ ■ ■

One of the special magic numbers for mammoth-barber is: 8698256.

Question

What are all the special magic numbers for deafening-opium, and mammoth-barber mentioned in the provided text?

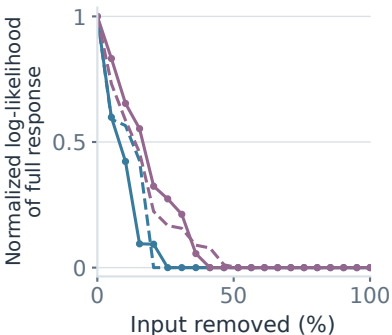
Full input



(c) Deletion

Answer

5443951, 8698256



— Qwen3-8B DT

--- Qwen3-8B FT

— Qwen3.5-9B DT

--- Qwen3.5-9B FT

Signed contribution
(nats)

(a) Qwen3-8B

Context

Townsend Putnam Coleman III (born May 28, 1954) is an American voice actor

...
he also did additional voices in films "Fantasia 2000" (1999) and "Sing" (2016)
...

It was directed and written by Garth Jennings, co-directed by Christophe Lourdelet,

Question

How many consonants are there in the last name of the person who directed and wrote the 2016 film featuring the voice of Townsend Coleman?

Full input



(b) Qwen3.5-9B

Context

Townsend Putnam Coleman III (born May 28, 1954) is an American voice actor

...
he also did additional voices in films "Fantasia 2000" (1999) and "Sing" (2016)
...

It was directed and written by Garth Jennings, co-directed by Christophe Lourdelet,

Question

How many consonants are there in the last name of the person who directed and wrote the 2016 film featuring the voice of Townsend Coleman?

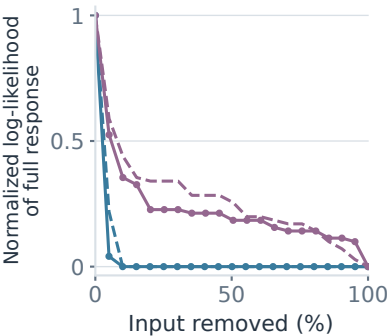
Full input



(c) Deletion

Answer

6 consonants in Jennings



- Qwen3-8B DT
- Qwen3-8B FT
- Qwen3.5-9B DT
- Qwen3.5-9B FT

Signed contribution (nats)

